



CS420 Sample Exam
SOLUTION

Instructor: Zhize LI

INSTRUCTIONS TO CANDIDATES

- Time allowed for this examination paper is **2 hours**.
- This examination paper contains **10** questions and comprises **15** pages inclusive of this instruction sheet. You are required to answer **ALL** questions within the spaces provided in the examination paper.
- This is a **CLOSED BOOK** examination.
- You are **ALLOWED** to bring only a single A4 sized cheat sheet.
- You are **ALLOWED** to use a calculator (which is not a smart device).
- Blank sheets for rough work are provided at the end of this paper.
- There shall be **NO ENQUIRY** during the exam. State any assumptions, if necessary, when you answer the questions.
- Methods, explanations, and justifications are important and may get you partial points.

Student Name: _____

Student ID: _____

Section: _____

Exam Venue: _____

Instructor Name: _____

Seat No

	Marks	Awarded
TOTAL	70	

Question 1.

A **fair** coin is tossed three times. The tosses are independent. Let H denote the number of Heads.

(a) What is $P(H \geq 1)$?

$$P(H \geq 1) = 1 - P(H=0) = 1 - 1/8 = 7/8$$

(b) Given that $H \geq 1$, what is $P(H=3 | H \geq 1)$?

$$P(H=3 | H \geq 1) = \frac{P(H=3 \cap H \geq 1)}{P(H \geq 1)} = \frac{1/8}{7/8} = 1/7$$

Question 2.

A factory has two machines:

- Machine A produces 40% of the products.
- Machine B produces 60% of the products.

It is known that: 5% of products produced by Machine A are defective; 2% of products produced by Machine B are defective.

Show your steps clearly.

(a) What is the probability that a randomly selected product is defective?

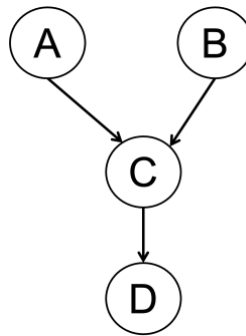
$$\begin{aligned}
 P(D=1) &= P(D=1, M=A) + P(D=1, M=B) \\
 &= P(M=A)P(D=1 | M=A) + P(M=B)P(D=1 | M=B) \\
 &= 0.4 \times 0.05 + 0.6 \times 0.02 \\
 &= 0.032
 \end{aligned}$$

(b) Given that a product is defective, what is the probability that it was produced by Machine A?

$$P(M=A | D=1) = \frac{P(M=A)P(D=1|M=A)}{P(D=1)} = 0.4 \times 0.05 / 0.032 = 0.625$$

Question 3.

Consider the Bayesian Network below:



(a) Factorize the joint distribution $P(A,B,C,D)$ according to the BN?

$$P(A,B,C,D) = P(A)P(B)P(C|A,B)P(D|C)$$

(b) For each statement, mark **True** or **False** according to the BN. No justification is required.

- 1) $A \perp B$
- 2) $A \perp B \mid C$
- 3) $A \perp D$
- 4) $A \perp D \mid C$

True
False
False
True

(c) Use variable elimination with **elimination order A, B, C** to compute the inference $P(D)$.

$$\begin{aligned}
 P(D) &= \sum_{A,B,C} P(A,B,C,D) \quad // \text{Marginalization} \\
 &= \sum_A \sum_B \sum_C P(A)P(B)P(C|A,B)P(D|C) \quad // \text{BN Factorization} \\
 &= \sum_B \sum_C P(B)P(D|C) \sum_A P(A)P(C|A,B) \quad // \text{Collect all factors dependent on } A \\
 &= \sum_B \sum_C P(B)P(D|C)\tau_1(B,C) \quad // \text{Sum out to eliminate } A \\
 &= \sum_C P(D|C) \sum_B P(B)\tau_1(B,C) \quad // \text{Collect all factors dependent on } B \\
 &= \sum_C P(D|C)\tau_2(C) \quad // \text{Sum out to eliminate } B \\
 &= \sum_C P(D|C)\tau_2(C) \quad // \text{Collect all factors dependent on } C \\
 &= \tau_3(D) \quad // \text{Sum out to eliminate } C
 \end{aligned}$$

Question 4.

You are building a Naïve Bayes classifier to classify an email into one of two classes: **Spam** or **Not Spam**.

You collected 100 emails:

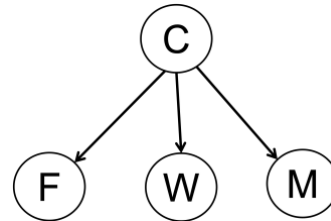
- 30 Spam
- 70 Not Spam

The table below shows how many emails in each class contain the words “free”, “win”, or “meeting”.

Words	Spam	Not Spam
“free”	15	7
“win”	18	7
“meeting”	6	35

- (a) Draw the Naïve Bayes model (i.e., the DAG) for this problem. For each node, specify the corresponding random variable and its conditional probability distribution (CPD).

$C \in \{\text{Spam, Not Spam}\}$ denote the class variable
 $F \in \{0, 1\}$ denote whether “free” appears
 $W \in \{0, 1\}$ denote whether “win” appears
 $M \in \{0, 1\}$ denote whether “meeting” appears



CPDs:

$$P(C=\text{Spam}) = 30/100 = 0.3$$

$$P(F=1|C=\text{Spam}) = 15/30 = 0.5, \quad P(F=1|C=\text{Not Spam}) = 7/70 = 0.1$$

$$P(W=1|C=\text{Spam}) = 18/30 = 0.6, \quad P(W=1|C=\text{Not Spam}) = 7/70 = 0.1$$

$$P(M=1|C=\text{Spam}) = 6/30 = 0.2, \quad P(M=1|C=\text{Not Spam}) = 35/70 = 0.5$$

- (b) Given an observed email that contains “free” and “win” but not “meeting”. What is the predicted class?

$$P(C, F, W, M) = P(C)P(F|C)P(W|C)P(M|C)$$

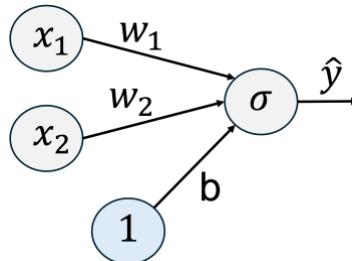
$$P(C=\text{Spam} | F=1, W=1, M=0) \propto P(C=\text{Spam}, F=1, W=1, M=0) = 0.3 \times 0.5 \times 0.6 \times 0.8 = 0.072$$

$$P(C=\text{Not Spam} | F=1, W=1, M=0) \propto P(C=\text{Not Spam}, F=1, W=1, M=0) = 0.7 \times 0.1 \times 0.1 \times 0.5 = 0.0035$$

Since $0.072 > 0.0035$, predicted class: **Spam**

Question 5.

Consider the following artificial neural network (ANN) that takes two inputs x_1, x_2 and outputs a real value $\hat{y}$. The ANN has no hidden layer, and the output layer uses a linear activation function $\sigma(z) = z$. The parameters are w_1, w_2 , and bias b .



- For a training sample i , the inputs are $x_{i,1}, x_{i,2}$, the true label is y_i , and the prediction is $\hat{y}_i$.
- (a) Suppose we use mean squared error (MSE): $L = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$. Write down explicitly the expression for the loss function for n data samples $\{(x_{i,1}, x_{i,2}, y_i)\}_{i=1}^n$, in terms of the parameters w_1, w_2 , and b .

The prediction for data sample i is:

$$\hat{y}_i = \sigma(w_1 x_{i,1} + w_2 x_{i,2} + b) = w_1 x_{i,1} + w_2 x_{i,2} + b.$$

Thus, the loss function is:

$$L(w_1, w_2, b) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \frac{1}{n} \sum_{i=1}^n (w_1 x_{i,1} + w_2 x_{i,2} + b - y_i)^2.$$

- (b) Compute the gradient of the loss function with respect to w_1, w_2 , and b .

$$\nabla L = \begin{pmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \frac{\partial L}{\partial b} \end{pmatrix} = \begin{pmatrix} \frac{2}{n} \sum_{i=1}^n (w_1 x_{i,1} + w_2 x_{i,2} + b - y_i) x_{i,1} \\ \frac{2}{n} \sum_{i=1}^n (w_1 x_{i,1} + w_2 x_{i,2} + b - y_i) x_{i,2} \\ \frac{2}{n} \sum_{i=1}^n (w_1 x_{i,1} + w_2 x_{i,2} + b - y_i) \end{pmatrix}$$

Question 6.

Consider the following 4x4x1 grayscale image:

$$X = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 4 & 1 & 2 & 0 \\ 1 & 3 & 1 & 2 \\ 0 & 1 & 2 & 4 \end{bmatrix}$$

You apply a convolutional layer with one filter

$$K = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

with stride = 2, no padding, and **bias** b = 1.

(a) Compute the output of the convolutional layer.

$$\begin{aligned} Y_{1,1} &= 1 \times 1 + 2 \times 0 + 4 \times 0 + 1 \times (-1) + 1 = 1 \\ Y_{1,2} &= 1 \times 1 + 1 \times 0 + 2 \times 0 + 0 \times (-1) + 1 = 2 \\ Y_{2,1} &= 1 \times 1 + 3 \times 0 + 0 \times 0 + 1 \times (-1) + 1 = 1 \\ Y_{2,2} &= 1 \times 1 + 2 \times 0 + 2 \times 0 + 4 \times (-1) + 1 = -2 \end{aligned}$$

The output is $Y = \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix}$

(b) The output of the convolutional layer is then passed through a 2x2 max-pooling layer. Compute the final output.

$$\max\{1, 2, 1, -2\} = 2$$

The final output is [2].

Question 7.

Consider an attention mechanism over three tokens: “**laptop**”, “**bag**”, “**it**”. Suppose the attention similarity scores (scaled dot product of Query and Key) for the token “**it**” are:

$$s_{it} = [3, 1, 2]$$

corresponding to [**laptop**, **bag**, **it**].

The Value vectors are:

$$v_{laptop} = [2, -1], \quad v_{bag} = [0, 2], \quad v_{it} = [1, 1]$$

(a) Compute the attention weights using softmax for “**it**”.

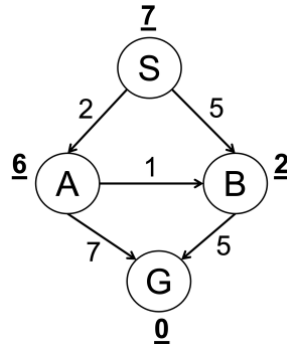
$$\begin{aligned} \text{The weights } w_{it} &= \text{softmax}(s_{it}) \\ &= \left[\frac{\exp(3)}{\exp(3)+\exp(1)+\exp(2)}, \frac{\exp(1)}{\exp(3)+\exp(1)+\exp(2)}, \frac{\exp(2)}{\exp(3)+\exp(1)+\exp(2)} \right] \\ &\approx [0.665, 0.090, 0.245] \end{aligned}$$

(b) Compute the attention output for “**it**”.

$$\begin{aligned} \text{The attention output for “it”} &: \\ w_{it}V &= 0.665v_{laptop} + 0.09v_{bag} + 0.245v_{it} \\ &= 0.665 \times [2, -1] + 0.09 \times [0, 2] + 0.245 \times [1, 1] \\ &= [1.575, -0.24] \end{aligned}$$

Question 8.

Using A* search to solve the following problem from the start state S to goal state G. The **bold and underlined** numbers are the heuristic estimates to the goal, and the numbers on the edges denote the step costs.



(a) Show the A* search process step by step. Report the final path and its cost.

A* search:

1. $S(7)$ **DQ:** $S(7)$ **ADD:** $A^S(8), B^S(7)$
2. $B^S(7), A^S(8)$ **DQ:** $B^S(7)$ **ADD:** $G^B(10)$
3. $A^S(8), G^B(10)$ **DQ:** $A^S(8)$ **ADD:** $B^A(5), G^A(9)$
4. $B^A(5), G^A(9)$ **DQ:** $B^A(5)$ **ADD:** $G^B(8)$
5. $G^B(8)$ **DQ:** $G^B(8)$

Retrieve the path: $G^B \leftarrow B^A \leftarrow A^S \leftarrow S$.

Path cost: 8.

(b) Is the given heuristic consistent?

Not consistent.

Question 9.

A delivery robot operates in a two-floor warehouse: **Ground** and **Upper** floor. At any time, the robot may or may not be **carrying a package**.

At each time step, the robot can choose:

- **Lift:**
 1. move to the **Upper** floor (if currently at **Ground**),
 2. pick up a package (if currently at **Upper** and **not carrying**),
 3. stay in the same state (otherwise).
- **Drop:**
 1. move to the **Ground** floor (if currently at **Upper**),
 2. deliver the package (if currently at **Ground** and **carrying**),
 3. stay in the same state (otherwise).

All transitions are **deterministic**.

Rewards:

- Successful delivery: **+4**
- The robot is carrying a package after the transition without delivery: **-1**
- Otherwise: **0**

(a) Formulate the MDP $\langle S, A, P, R \rangle$ for the above setting.

States (List all states in the MDP):

$S = \{(\text{Ground, Carrying}), (\text{Ground, Non-carrying}), (\text{Upper, Carrying}), (\text{Upper, Non-carrying})\}$

Actions (List all actions in the MDP):

$A = \{\text{Lift, Drop}\}$

Transitions (Since the state transitions are deterministic, for each state-action pair, indicate the resulting next state s' in the table below. You may add rows if necessary.):

State s	Action a	Next state s'
(Ground, Carrying)	Lift	(Upper, Carrying)
(Ground, Carrying)	Drop	(Ground, Non-carrying)
(Ground, Non-carrying)	Lift	(Upper, Non-carrying)
(Ground, Non-carrying)	Drop	(Ground, Non-carrying)
(Upper, Carrying)	Lift	(Upper, Carrying)
(Upper, Carrying)	Drop	(Ground, Carrying)
(Upper, Non-carrying)	Lift	(Upper, Carrying)
(Upper, Non-carrying)	Drop	(Ground, Non-carrying)

Rewards (Write the reward function values in the table below. You may add rows if necessary.):

State s	Action a	Reward $R(s, a, s')$
(Ground, Carrying)	Lift	-1
(Ground, Carrying)	Drop	+4
(Ground, Non-carrying)	Lift	0
(Ground, Non-carrying)	Drop	0
(Upper, Carrying)	Lift	-1
(Upper, Carrying)	Drop	-1
(Upper, Non-carrying)	Lift	-1
(Upper, Non-carrying)	Drop	0

- (b) Use Value Iteration (VI) to compute $V^1(s)$ for all states. Let discount factor $\gamma = 0.9$ and initialize $V^0(s) = 0$ for all states.

Since transitions are deterministic, we have:

$$Q^1(s, a) = R(s, a, s') + \gamma \cdot V^0(s') = R(s, a, s')$$

$$V^1(s) = \max_a Q^1(s, a)$$

$$Q^1((\text{Ground, Carrying}), \text{Lift}) = -1, \quad Q^1((\text{Ground, Carrying}), \text{Drop}) = 4$$

$$V^1((\text{Ground, Carrying})) = 4$$

$$Q^1((\text{Ground, Non-carrying}), \text{Lift}) = 0, \quad Q^1((\text{Ground, Non-carrying}), \text{Drop}) = 0$$

$$V^1((\text{Ground, Non-carrying})) = 0$$

$$Q^1((\text{Upper, Carrying}), \text{Lift}) = -1, \quad Q^1((\text{Upper, Carrying}), \text{Drop}) = -1$$

$$V^1((\text{Upper, Carrying})) = -1$$

$$Q^1((\text{Upper, Non-carrying}), \text{Lift}) = -1, \quad Q^1((\text{Upper, Non-carrying}), \text{Drop}) = 0$$

$$V^1((\text{Upper, Non-carrying})) = 0$$

Question 10.

Consider the following 2x3 grid world:

S1	S2	S3
S4	S5	G

The states are $S = \{S1, S2, S3, S4, S5, G\}$, where G is the terminal state.

The actions are four directions: $A = \{\text{Up, Down, Left, Right}\}$.

The agent uses **Q-learning** with

- stepsize/learning rate $\alpha = 0.1$,
- discount factor $\gamma = 0.9$,
- all Q-values initialized to 0.

Suppose that the agent collects the following **three experiences (one episode)**, where each experience is of the form (s, a, r, s') :

Experience #	s	a	r	s'
1	S1	Right	-1	S2
2	S2	Right	-1	S3
3	S3	Down	5	G

(a) Process the experiences in the given order. Write the **updated Q-values after this episode**.

To save space, write only the **updated entries**. You may add more rows if necessary.

State s	Action a	$Q(s, a)$
S1	Right	-0.1
S2	Right	-0.1
S3	Down	0.5

Q-learning update for an experience (s, a, r, s') :

$$Q^{\text{new}}(s, a) = Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

[rough paper]

[rough paper]

[rough paper]